

FIITJEE
ALL INDIA TEST SERIES
JEE (Advanced)-2022
OPEN TEST – II
PAPER –1
TEST DATE: 20-03-2022

Time Allotted: 3 Hours

Maximum Marks: 180

General Instructions:

- The test consists of total 57 questions.
- Each subject (PCM) has 19 questions.
- This question paper contains **Three Parts**.
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Three Sections: Section-A, Section-B & Section-C**.
Section – A (01 – 04, 20 – 23, 39 – 42): This section contains **TWELVE (12)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.
Section – A (05 – 10, 24 – 29, 43 – 48): This section contains **EIGHTEEN (18)** questions. Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
Section – B (11 – 13, 30 – 32, 49 – 51): This section contains **NINE (09)** questions. The answer to each question is a **NON-NEGATIVE INTEGER**.
Section – C (14 – 19, 33 – 38, 52 – 57): This section contains **NINE (09)** question stems. There are **TWO (02)** questions corresponding to each question stem. The answer to each question is a **NUMERICAL VALUE**. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

MARKING SCHEME

Section – A (Single Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+3	If ONLY the correct option is chosen.
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	-1	In all other cases.

Section – A (One or More than One Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If only (all) the correct option(s) is (are) chosen;
Partial Marks	:	+3	If all the four options are correct but ONLY three options are chosen;
Partial marks	:	+2	if three or more options are correct but ONLY two options are chosen and both of which are correct;
Partial Marks	:	+1	If two or more options are correct but ONLY one option is chosen and it is a correct option;
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	-2	In all other cases.

Section – B: Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If ONLY the correct integer is entered;
Zero Marks	:	0	In all other cases.

Section – C: Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+2	If ONLY the correct numerical value is entered at the designated place;
Zero Marks	:	0	In all other cases.

Physics

PART – I

Section – A (Maximum Marks: 12)

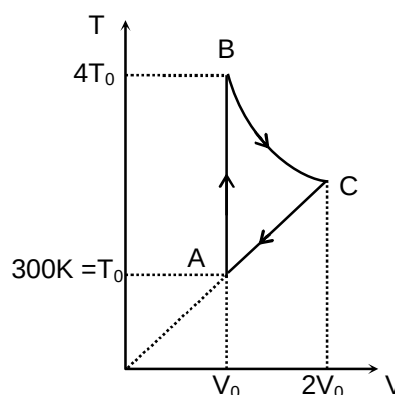
This section contains **FOUR (04)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.

1. A uniform rope of length $\ell = 12$ m is suspended between two fixed nails A and B that are in the same horizontal level. The rope makes an angle $\theta = 37^\circ$ with the horizontal at the nails. The radius of curvature of the rope at the lowest point is



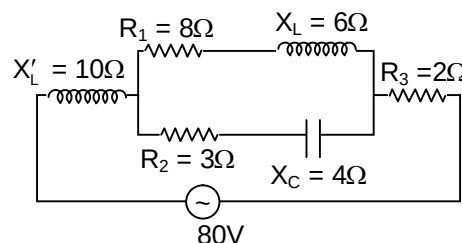
- (A) 2 m
(B) 4 m
(C) 6 m
(D) 8 m

2. One mole of an ideal monoatomic gas is taken through a cyclic process ABCA as shown in the T-V diagram. In the process BC, $TV = \text{constant}$. The efficiency of the cycle in percentage is



- (A) 11.2
(B) 22.2
(C) 33.2
(D) 44.2

3. Consider the circuit shown in the figure. The current supplied by the ac voltage source is



- (A) 4 amp
(B) 8 amp
(C) 10 amp
(D) 16 amp

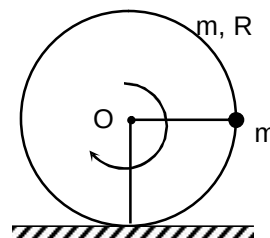
4. A metallic surface is illuminated alternatively with electromagnetic radiations of wavelengths $3100\text{\AA}$ and $4000\text{\AA}$. It is observed that the maximum speeds of the photoelectrons under these illuminations are in the ratio 2 : 1. The work function of the metal is

- (A) 2 eV
(B) 2.4 eV
(C) 2.8 eV
(D) 3 eV

Section – A (Maximum Marks: 24)

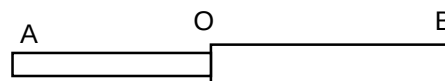
This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

5. A particle of mass m is rigidly attached to the circumference of a uniform circular ring of same mass m and radius R which is placed on a rough horizontal surface as shown in the figure. When the ring is released from rest from the initial position shown in the figure, it starts rolling without slipping on the rough horizontal surface. At the moment when the ring has rotated by an angle $\theta = 30^\circ$, choose the correct option(s).

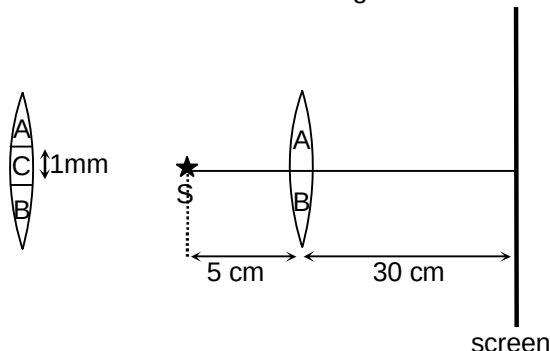


- (A) The angular velocity of the ring is $\sqrt{\frac{g}{3R}}$
- (B) The angular acceleration of the ring is $\frac{2g}{3\sqrt{3}R}$
- (C) The frictional force acting on the ring is $\frac{mg}{\sqrt{3}}$
- (D) The frictional force acting on the ring is $\frac{mg}{2\sqrt{3}}$

6. Two strings of mass per unit length $\mu_1 = 0.09 \text{ kg/m}$ and $\mu_2 = 0.04 \text{ kg/m}$ are joined together and stretched between two rigid supports as shown in the figure. A sinusoidal wave of amplitude 2.5 cm is sent from the end A. Then choose the correct option(s).

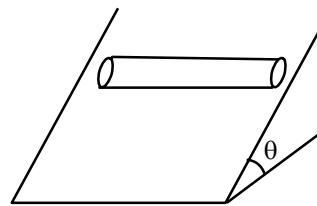


- (A) The amplitude of the reflected wave from the joint O is 0.5 cm.
- (B) The amplitude of the transmitted wave from the joint O is 3 cm.
- (C) The percentage of the incident power transmitted from the joint O is 96%.
- (D) The percentage of the incident power reflected from the joint O is 20%.
7. A convex lens of focal length 10 cm is cut along the diameter into two identical halves A and B and in the process a central layer C of the lens of thickness 1 mm is lost. Then the two halves A and B are put together to form a composite lens. A point source of light emitting wavelength $\lambda = 5000 \text{ \AA}$ is placed at a distance 5 cm in front of this composite lens and a screen is placed behind the lens at a distance 30 cm from it as shown in the figure. Then choose the correct option(s).



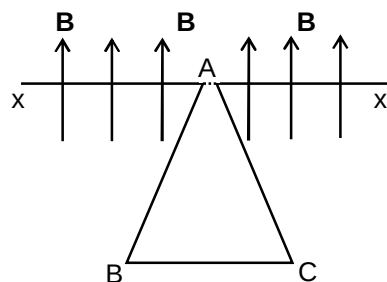
- (A) The fringe width of the interference pattern obtained on the screen is 0.2 mm.
- (B) The fringe width of the interference pattern obtained on the screen is 0.6 mm.
- (C) The number of interference bands observed on the screen is 15.
- (D) The number of interference bands observed on the screen is 12.

8. Charge Q is uniformly distributed over the curved surface of a non-conducting long hollow cylinder of mass m , length ℓ and radius R which is placed on a rough inclined plane of inclination θ as shown in the figure. The cylinder starts rolling without slipping down the rough inclined plane. Then choose the correct option(s). (Take $\frac{\mu_0 Q^2}{\pi \ell} = m$)



- (A) The acceleration of the cylinder is $\frac{4g \sin \theta}{9}$
 (B) The acceleration of the cylinder is $\frac{2g \sin \theta}{9}$
 (C) The frictional force acting on the cylinder is $\frac{2mg \sin \theta}{9}$
 (D) The frictional force acting on the cylinder is $\frac{5mg \sin \theta}{9}$

9. A uniform conducting wire of mass $3m$ and length 3ℓ is bent into the shape of an equilateral triangular loop ABC and suspended in a vertical plane from a fixed horizontal axis xx' as shown in the figure. The loop is free to rotate about the fixed horizontal axis xx' . Each side of the triangular loop is having mass m and length ℓ . A uniform vertical upward magnetic field B is existing in the region. The charge Q is passed through the triangular loop almost instantaneously in the initial vertical position. If the maximum deflection of the triangular loop from vertical plane is $\theta = \frac{\pi}{2}$, then choose the correct option(s).



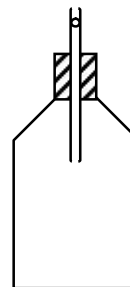
- (A) The initial angular velocity acquired by the loop is $\sqrt{\frac{2\sqrt{3}g}{5\ell}}$
 (B) The initial angular velocity acquired by the loop is $\sqrt{\frac{8\sqrt{3}g}{5\ell}}$
 (C) The value of magnetic field B is $\frac{m}{Q} \sqrt{\frac{40g}{\sqrt{3}\ell}}$
 (D) The value of magnetic field B is $\frac{m}{Q} \sqrt{\frac{10g}{\sqrt{3}\ell}}$

10. The pitch of a screw gauge is 1 mm and its cap is divided into 100 divisions. When nothing is placed between its studs, the zero of the circular scale is lying 6 divisions above the reference line. When a wire is placed between its studs, the main scale reading is 3 divisions and 58th division of circular scale coincides with the reference line. Then choose the correct option(s).
 (A) The least count of the screw gauge is 0.01 mm.
 (B) The zero error in the screw gauge is - 0.06 mm
 (C) The diameter of the wire is 3.52 mm
 (D) The diameter of the wire is 3.64 mm.

Section – B (Maximum Marks: 12)

This section contains **THREE (03)** questions. The answer to each question is a **NON-NEGATIVE INTEGER**.

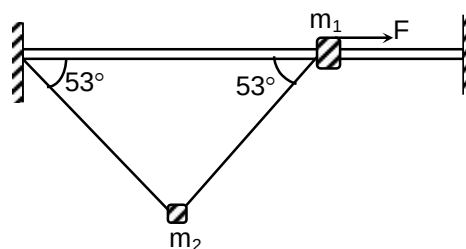
11. A thin circular plate of radius R is held closing the opening of a thin hemispherical shell of the same radius R . Both the bodies are made of insulating materials and having uniform surface charge density σ each. The plate is released keeping the hemispherical shell fixed. If the maximum kinetic energy acquired by the plate is $n \left(\frac{\sigma^2 \pi R^3}{32 \epsilon_0} \right)$. The value of n is.....
12. Opening of a glass bottle containing an ideal monoatomic gas is tightly closed with a cork fitted with a thin frictionless glass tube of cross-section area S . Inner volume of the tube is negligibly small as compared to volume V of the bottle. A small glass marble of mass m , when dropped in the tube, it starts oscillating up and down in the tube without friction. The marble exactly fits inside the tube so that there is no leakage of air. The atmospheric pressure is P_0 and consider the glass, the cork and the material of the marble as perfect insulator of heat. If the time period of oscillations of the marble is $2\pi \sqrt{\frac{kmV}{15S(P_0S + mg)}}$. The value of k is.....
13. A longitudinal standing wave $y = A \sin kx \cos \omega t$ is maintained in a homogeneous rod with cross-sectional area S , length ℓ and density ρ . If the elastic potential energy stored in the rod at time $t = \frac{\pi}{6\omega}$ is $\alpha \left(\frac{\rho \omega^2 A^2 S \ell}{32} \right)$. The value of α is

**Section – C (Maximum Marks: 12)**

This section contains **THREE (03)** question stems. There are **TWO (02)** questions corresponding to each question stem. The answer to each question is a **NUMERICAL VALUE**. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

Question Stem for Question Nos. 14 and 15**Question Stem**

A sleeve of mass $m_1 = 5$ kg can slide on a smooth horizontal rod. A block of mass $m_2 = 2$ kg is suspended at the mid-point of a light inextensible string of length $\ell = 2.3$ m, one end of which is attached to the sleeve and the other end is attached to one end of the rod. Initially the sleeve is held at rest and the angle between the rod and a string segment is $\theta_0 = 53^\circ$ as shown in the figure. A constant horizontal force $F = 20$ N is applied on the sleeve. When the angle between the rod and the string segment becomes $\theta = 37^\circ$, the speeds of the sleeve and the block are v_1 m/s and v_2 m/s respectively. (Take $g = 10$ m/s²)



14. The value of v_1 is
15. The value of v_2 is

Question Stem for Question Nos. 16 and 17

Question Stem

In a setup of displacement method experiment, distance between the screen and a light source is 100 cm and the lens used has a small aperture. By moving the lens between the source and the screen, sharp images are obtained on the screen for two different positions of the lens separated by a distance x cm. The ratio of sizes of these two images is 9 : 1. The focal length of the lens used is f cm.

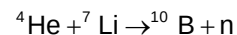
16. The value of x is

17. The value of f is

Question Stem for Question Nos. 18 and 19

Question Stem

When an α -particle with kinetic energy $k = 14$ MeV collides with a stationary lithium nucleus, the following nuclear reaction takes place.



The Q-value of this reaction is $E = -2.9$ MeV and the neutron is outgoing at right angle to the incoming direction of the α -particle. If kinetic energy of the neutron and the boron nucleus are k_1 MeV and k_2 MeV respectively.

18. The value of k_1 is

19. The value of k_2 is

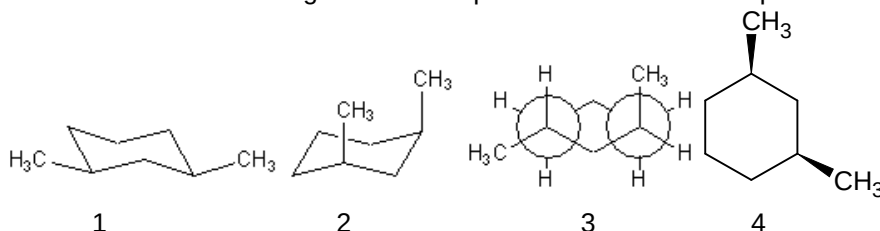
Chemistry

PART – II

Section – A (Maximum Marks: 12)

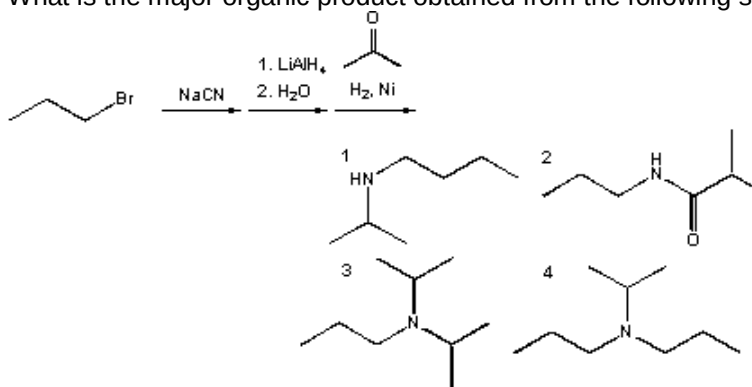
This section contains **FOUR (04)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.

20. Which one of the following structures represents a different compound from the other three?



- (A) 1
(B) 2
(C) 4
(D) 3

21. What is the major organic product obtained from the following sequence of reactions?



- (A) 1
(B) 4
(C) 3
(D) 2

22. Which is / are correct statements (s) about P_4O_6 and P_4O_{10} ?

- On hydrolysis Both form oxyacid's H_3PO_3 and H_3PO_4 respectively.
- In P_4O_6 each P is joined to four O and in P_4O_{10} each P is joined to six O atoms
- In P_4O_6 each P is joined to three O and in P_4O_{10} each P is joined to four O atoms
- In P_4O_6 and in P_4O_{10} each P is SP^3 hybridized.

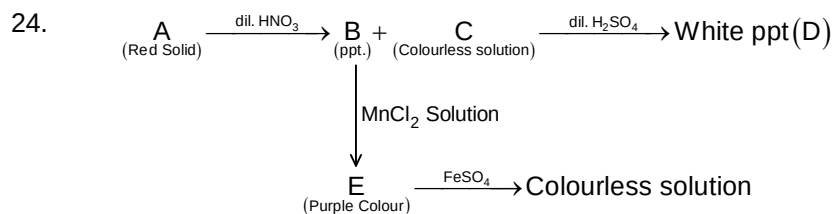
- (A) 1, 3, 4 only
(B) 1, 4 only
(C) 1, 3 only
(D) 2, 3, 4 only

23. How many mL of 1.00 M NaOH must be added to 100 ml of 0.1 M H_3PO_4 solution to obtain a phosphate buffer solution with pH of about 8.2? (The pK values of H_3PO_4 are $pK_1 = 2.1$, $pK_2 = 8.2$, $pK_3 = 12$)

- (A) 10.0 ML
(B) 15.0 ML
(C) 20.0 ML
(D) 30 ML

Section – A (Maximum Marks: 24)

This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).



- (A) Compound A is Pb_3O_4
 (B) Compound D is PbSO_4
 (C) Compound B is PbO_2
 (D) Compound E is MnO_4^-

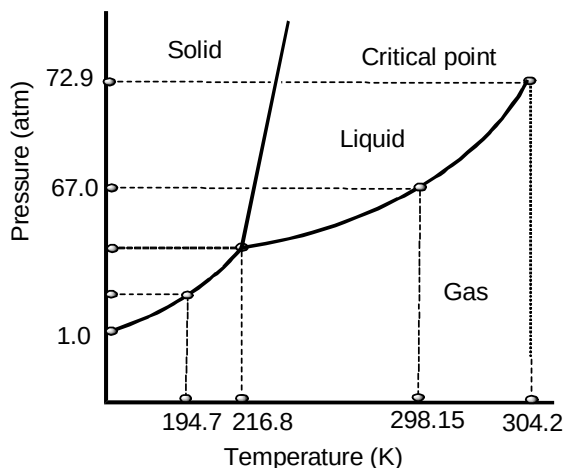
25. PCl_3 gives the same TYPE of products on hydrolysis as

- (A) NCl_3 gives
 (B) BCl_3 gives
 (C) NF_3 gives
 (D) BiCl_3 gives

26. The correct statement about the interhalogen compound ICl_3 is:

- (A) It exists as dimer.
 (B) Geometry around the iodine is tetrahedral in solid state.
 (C) It decomposes as ICl and Cl_2 in gas phase.
 (D) Liquid ICl_3 conducts electricity.

27. Consider the following phase diagram of CO_2 (not to scale). At equilibrium, the correct statement is:



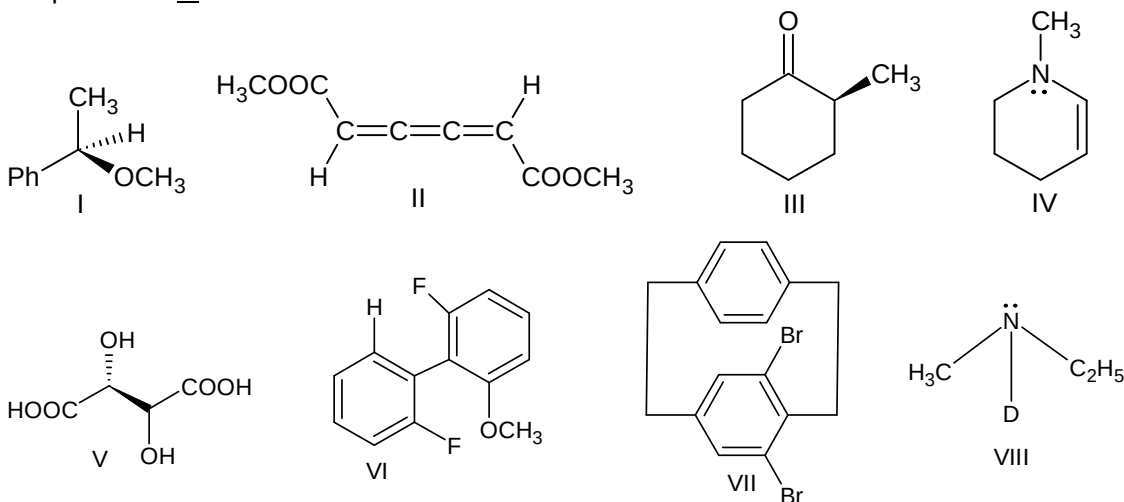
- (A) At 200 K, on increasing the pressure from 1 to 50 atm, CO_2 gas condenses to liquid.
 (B) It is not possible to obtain liquid CO_2 from gaseous CO_2 below 5.11 atm.
 (C) Both liquid and gas phase of CO_2 coexist at 298.15 K and 67 atm.
 (D) With increasing pressure, the melting point of solid CO_2 increases.

28. The rate of the following redox reaction is slowest when **X** is
 $[\text{Co}^{\text{III}}(\text{NH}_3)_5\text{X}]^{3+/2+} + [\text{Cr}^{\text{II}}(\text{H}_2\text{O})_6]^{2+} \rightarrow [\text{Co}^{\text{II}}(\text{NH}_3)_5(\text{H}_2\text{O})]^{2+} + [\text{Cr}^{\text{III}}(\text{H}_2\text{O})_5\text{X}]^{3+/2+}$
 (A) H_2O
 (B) NH_3
 (C) Cl^-
 (D) N_3^-
29. Micelle formation is accompanied by the
 (A) decrease in overall entropy due to ordering.
 (B) mostly due to increase in solvent entropy increase in overall entropy.
 (C) increase in overall entropy mostly due to increase in solute entropy.
 (D) increase in overall entropy and decrease in enthalpy.

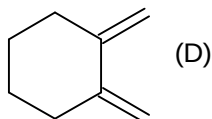
Section – B (Maximum Marks: 12)

This section contains **THREE (03)** questions. The answer to each question is a **NON-NEGATIVE INTEGER**.

30. Among the following compounds, the number of compounds that exhibit optical activity at room temperature is__.



31. When the following diene **D** undergo Diels-Alder reaction with methyl acrylate is form a major product **P**, which on allylic MONO bromination with NBS produces X number of possible products (without rearrangements). Value of X is_____?



32. The vapor pressure of toluene is 0.13 atm at 25 °C. If 6 g of a compound X is dissolved in 46 g of toluene at 25°C, the vapor pressure drops to 0.12 atm. The molar mass (in g/mol) of the compound X is----

Section – C (Maximum Marks: 12)

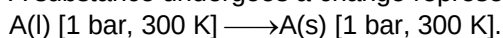
This section contains **THREE (03)** question stems. There are **TWO (02)** questions corresponding to each question stem. The answer to each question is a **NUMERICAL VALUE**. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

Question Stem for Question Nos. 33 and 34

Question Stem

Question Stem For the following reaction scheme

A substance undergoes a change represented as shown



From the given information, if the magnitude of change in internal energy (in Joules) when 1 moles of A(l) solidifies is X (J/mol).

Information :

- (i) Standard melting point of A(s) is 300 K.
- (ii) Latent heat of fusion of A is 0.01 kJ/gm.
- (iii) Specific volume of A(s) is 100 ml/g and A(l) is 120 ml/gm.
- (iv) Molar mass of A is 50 gm / mole.

33. The value of magnitude of ΔH (J/mol). is ____.

34. The value of X is ____

Question Stem for Question Nos. 35 and 36

Question Stem

A certain salt (X) gives the following test.

- (i) Its aqueous solution is alkaline to litmus.
- (ii) On strong heating it swells to give a glassy material.
- (iii) When concentrated sulphuric acid is added to a hot concentrated solution of (X), white crystals of a weak acid (Y) separate out.

35. The Percentage s-character in central atom of Y is ____?

36. Y on strong heating gives a white color solid (Z), calculate the percentage of oxygen by mass in Z ____?

Question Stem for Question Nos. 37 and 38

Question Stem

The hydrogen-like radial wave function of the 3s orbital is given as

$$R_{3,0} = \frac{1}{9\sqrt{3}} \left(\frac{Z}{a_0} \right)^{3/2} \left(6 - 2\rho + \frac{\rho^2}{9} \right) e^{-\rho/6}$$

where $\rho = 2Zr/a_0$; Z = atomic number; r = distance from the nucleus and a_0 = Bohr radius.

Positions of the radial nodes (in units of a_0) of the 3s orbital are at x and y .

37. Calculate distance between x and y (in units of a_0) for Li^{+2} ion ____?

38. Calculate ratio of y to x (in units of a_0) for Be^{+3} ion ____?

Mathematics

PART – III

Section – A (Maximum Marks: 12)

This section contains **FOUR (04)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.

39. If $\begin{bmatrix} 3 & 2 \\ 4 & 3 \end{bmatrix}^{2022} + \begin{bmatrix} 3 & -2 \\ -4 & 3 \end{bmatrix}^{2022} = (a^{2022} + b^{2022})I$ where $a, b \in \mathbb{R}$, then value of $a^2b + ab^2$ is
 (A) 7
 (B) 1
 (C) -7
 (D) 6
40. The value of $\lim_{n \rightarrow \infty} \left((n+2)!^{\frac{1}{n+2}} - (n)!^{\frac{1}{n}} \right)$, given that it exists finitely, is
 (A) $2e$
 (B) $\frac{e}{2}$
 (C) $\frac{2}{e}$
 (D) $\frac{1}{2e}$
41. Let $f : \mathbb{R} \rightarrow [0, 3]$ be a differentiable function such that $f(x)^2 f'(x) = 3f(3x)^2 f'(3x)$ and $\lim_{x \rightarrow \infty} f(x) = 2$, then the number of roots of the equation $f(x) = \cos x + 2x - (\cos^2 x + x^2)$ is
 (A) 0
 (B) 1
 (C) 2
 (D) none of these
42. z_1, z_2, z_3 are three unimodular complex numbers which are also roots of the equation $z^3 + az^2 + bz + 1 = 0$ (where a, b are complex numbers). If tangents are drawn to $|z| = 1$ at points z_1, z_2, z_3 which intersect pair wise at points $\omega_1, \omega_2, \omega_3$, then $\omega_1, \omega_2, \omega_3$ are roots of the equation
 (A) $(ab - 1)z^3 + 2(b^2 - a)z^2 + 8bz + 8 = 0$
 (B) $(ab - 1)z^3 + 2(b^2 + a)z^2 - 8bz + 8 = 0$
 (C) $(ab - 1)z^3 - 2(b^2 + a)z^2 + 8bz + 8 = 0$
 (D) $(ab - 1)z^3 + 2(b^2 + a)z^2 + 8bz + 8 = 0$

Section – A (Maximum Marks: 24)

This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

43. In a triangle ABC points D, E lie on sides AC and AB respectively. M and N mid points of side BD and CE respectively. If Δ_1 is area of triangle AMN, Δ_2 be the area of triangle ADE, Δ be area of triangle ABC, Δ_3 be area of quadrilateral BCDE, Δ_4 be area of quadrilateral MNDE. Which of the following is always true?
 (A) $4\Delta_1 = \Delta_3$
 (B) $4\Delta_1 + \Delta_2 = \Delta$
 (C) $\Delta_1 = \Delta_4$
 (D) $4\Delta_4 = \Delta_2 + \Delta$

44. Tangents are drawn from $\left(-\frac{\pi}{2}, 0\right)$ to $f: [0, \infty) \rightarrow \mathbb{R}; y = \sin x$ and $(a_i, \sin a_i)$ are the points of contact of these tangents for $i = 1, 2, 3, \dots$ and $a_{i+1} > a_i$. Choose the **correct** option(s)
- (A) $a_n + \frac{\pi}{2} = \tan a_n$
 (B) $a_{n+1} - a_n < \pi$
 (C) $a_{n+2} - a_n > 2\pi$
 (D) $\sin a_{2n} - \sin a_{2n-1} < 0$
45. P is a point on ellipse $\frac{x^2}{36} + \frac{y^2}{20} = 1$ in first quadrant. $F_1(\alpha, 0)$ and $F_2(\beta, 0)$ are two foci of the ellipse ($\alpha < \beta$). Line PF_1 intersect circle $x^2 + y^2 - 6y + 8 = 0$ in Q. Choose the **correct** option(s)
- (A) the maximum value of $PQ + PF_2$, as P moves on the ellipse in first quadrant, is 8.
 (B) the minimum value of $PQ + PF_2$, as P moves on the ellipse in first quadrant, is 6.
 (C) if the line PF_1 is at maximum distance from $(0, 3)$ then the value of $PQ + PF_2$ is $12 - \sqrt{24}$
 (D) the maximum value of $PQ + PF_2$, as P moves on the ellipse in first quadrant, is 12
46. A fair die is rolled. A is the event that odd number appears. B is the event that number appearing on the die is factor of a natural number m, where $1 \leq m \leq 60$. If A and B are independent events choose the **correct** option(s)
- (A) sum of all possible values of m is 368
 (B) sum of all possible values of m is 348
 (C) number of such possible values of m is 11
 (D) number of such possible values of m is 12
47. ABCD is a tetrahedron with $\triangle BCD$ being equilateral, having each side equal to 12 units. H is foot of perpendicular from A to $\triangle BCD$ and it lies inside the triangle if area of $\triangle CDH$ is three times the area of $\triangle BCH$ and area of $\triangle DBH$ is two times the area of $\triangle BCH$, $AH = 3$ and M is midpoint of side BD. Choose the **correct** option(s)
- (A) perpendicular distance of point H from side BC is $\sqrt{3}$
 (B) perpendicular distance of point H from side BD is $2\sqrt{3}$
 (C) perpendicular distance of point A from line CM is $\sqrt{13}$
 (D) perpendicular distance of point A from line CM is $\sqrt{15}$
48. $f: [0, 1] \rightarrow \mathbb{R}$ be a function such that $f(x)^2 = 1$ and $f(x)$ is discontinuous at exactly two point and $\int_0^1 f(x) dx = 0$. Let $I = \int_0^1 x f(x) dx$, choose the **incorrect** option(s)
- (A) maximum value of I is $\frac{1}{4}$
 (B) minimum value of I is $-\frac{3}{4}$
 (C) when I is maximum then number of such function $f(x)$ is 8
 (D) when I is minimum then number of such function $f(x)$ is 4

Section – B (Maximum Marks: 12)

This section contains **THREE (03)** questions. The answer to each question is a **NON-NEGATIVE INTEGER**.

49. If m is the minimum value of $f(x)$, where $f(x) = x^{3/2} + \frac{1}{8x^{3/2}} - \frac{17}{4}\left(x + \frac{1}{4x}\right)$, $x > 0$, then the value of $100 + 2m$ is
50. Let $p = \sum_{r=1}^{2021} (-1)^{r-1} \frac{{}^{2021}C_r}{2^{r+1} - 1}$ and $q = \sum_{r=1}^{\infty} \frac{(2^r - 1)^{2021}}{2^{2022r}}$, then the value of $p + q$ is
51. The function $f(x)$ defined for all $x > a$ and the $g(x)$ is four degree polynomial with leading coefficient -1 such that $(x - a)f(x) = g(x)$, $x > a$. $f(x)$ has local maximum value M at $x = \alpha$ and $x = \beta$ ($\alpha \neq \beta$, $M > 0$). The number of points where $f(x)$ has a local maxima or a local minima is more than the number of points where the function $g(x)$ has the local maxima or local minima. If $\beta - \alpha = 6\sqrt{3}$, then find minimum value of M

Section – C (Maximum Marks: 12)

This section contains **THREE (03)** question stems. There are **TWO (02)** questions corresponding to each question stem. The answer to each question is a **NUMERICAL VALUE**. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

Question Stem for Question Nos. 52 and 53**Question Stem**

The quadratic equation $x^2 - (a + 5)x + 2a = 0$, $a \in \mathbb{I}$ has rational roots α and β

52. The maximum value of $\alpha^2 + \beta^2$ is
53. The maximum value of $\alpha^2 + \beta^2$ for which $|\alpha - \beta|$ is minimum is

Question Stem for Question Nos. 54 and 55**Question Stem**

A box has 14 balls, 7 black balls numbered from 1 to 7 and another 7 white balls numbered from 1 to 7. A person picks 2 balls at a time and repeats the process till box is empty

54. Let n be the number of ways of drawing the balls when every time he picks balls of both the colours and difference of number appearing on them is at most one. The value of $\frac{n}{5!}$ is
55. Let n be the total number of ways of drawing the balls when every time he picks the balls such that difference of number appearing on them is at most one. The value of $\frac{n}{6!}$ is
(balls can be of any colour)

Question Stem for Question Nos. 56 and 57

Question Stem

AB is a focal chord of the parabola $y^2 = 4x$. C is a point on parabola such that CF is perpendicular to AB, F being the focus of parabola. A line through C, parallel to axis of parabola cuts AB at P. $f(x, y) = 0$ is locus of point P as AB varies. Let $y = g(x)$ be image of $f(x, y) = 0$ in the line $y = x$.

56. The number of integral values which does not belong to range of $y = g(x)$ is

57. Let (x_0, y_0) be point of contact of tangent on $y = g(x)$ from origin, the value of $\frac{(x_0^2 + 4)}{\sqrt{3}}$ is